

Manual Change History (4021674) QST30 CM850 Power Generation Interface Engine Electronic Control System Troubleshooting and Repair Manual

Date	Not Available
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Procedure Number	4021674-toc (/qs3/pubsys2/xml/en/manual/4021674/4021674-toc.html)
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Update Reason	

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Date	03-Apr-2003
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Date	09-Oct-2003
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Date	27-Jun-2006
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Date	13-Dec-2007
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Update Reason	

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Update Reason	
Date	14-Dec-2007
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Procedure Title	Service Tools
Update Reason	

Date	14-Dec-2007
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Update Reason	
Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Coolant Pressure Sensor
Update Reason	
Date	14-Dec-2007
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Procedure Title	Engine Coolant Level Sensor
Update Reason	

Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Update Reason	
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Section	Section 19 – Electronic Engine Controls – Group 19
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Update Reason	
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Update Reason	
Date	14-Dec-2007
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Update Reason	

Date	14-Dec-2007
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Update Reason	
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Update Reason	
Date	14-Dec-2007
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Update Reason	
Date	14-Dec-2007
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Update Reason	

Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Intake Manifold Pressure Sensor
Update Reason	
Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Engine Oil Pressure Sensor
Update Reason	
Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Engine Oil Temperature Sensor
Update Reason	
Date	14-Dec-2007
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Update Reason	

Date	14-Dec-2007
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Procedure Title	Engine Speed Sensor Circuit
Update Reason	
Date	14-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Burn Valve Solenoid, Centinel™ Oil
Update Reason	
Date	19-Dec-2007
Section	Section 19 – Electronic Engine Controls – Group 19
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Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	
Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
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Procedure Title	Engine Timing Actuator Driver Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
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Procedure Title	Engine Timing Actuator Driver Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
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Procedure Title	Engine Magnetic Speed/Position Lost Both of Two Signals – Data Erratic, Intermittent, or Incorrect
Update Reason	

Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc141 (/qs3/pubsys2/xml/en/procedures/60/60-fc141.html)
Procedure Title	Engine Oil Rifle Pressure 1 Sensor Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
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Update Reason	

Date	08-Jan-2008
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Procedure Title	Engine Coolant Temperature 1 Sensor Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	08-Jan-2008
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Procedure Title	Engine Coolant Temperature 1 Sensor Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	08-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
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Update Reason	

Date	08-Jan-2008
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Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Procedure Number	60-fc156 (/qs3/pubsys2/xml/en/procedures/60/60-fc156.html)
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Update Reason	

Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Date	08-Jan-2008
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Date	08-Jan-2008
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Date	08-Jan-2008
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Update Reason	
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Update Reason	

Date	08-Jan-2008
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Update Reason	

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Update Reason	

Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Date	08-Jan-2008
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Date	08-Jan-2008
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Update Reason	

Date	08-Jan-2008
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Date	08-Jan-2008
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Update Reason	
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Update Reason	

Date	09-Jan-2008
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Update Reason	
Date	09-Jan-2008
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Update Reason	
Date	09-Jan-2008
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Procedure Title	CM850 or CM552 Choice
Update Reason	
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Update Reason	

Date	09-Jan-2008
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Procedure Title	Engine Control Module – Out of Calibration
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc343 (/qs3/pubsys2/xml/en/procedures/60/60-fc343.html)
Procedure Title	Engine Control Module Warning Internal Hardware Failure – Bad Intelligent Device or Component
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc346 (/qs3/pubsys2/xml/en/procedures/60/60-fc346.html)
Procedure Title	Engine Control Module Warning Software Error – Bad Intelligent Device or Component
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc352 (/qs3/pubsys2/xml/en/procedures/60/60-fc352.html)
Procedure Title	Sensor Supply 1 Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc386 (/qs3/pubsys2/xml/en/procedures/60/60-fc386.html)
Procedure Title	Sensor Supply 1 Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc415 (/qs3/pubsys2/xml/en/procedures/60/60-fc415.html)
Procedure Title	Engine Oil Rifle Pressure 1 – Data Valid but Below Normal Operational Range – Most Severe Level
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc421 (/qs3/pubsys2/xml/en/procedures/60/60-fc421.html)
Procedure Title	Engine Oil Temperature – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc441-choice (/qs3/pubsys2/xml/en/procedures/60/60-fc441-choice.html)
Procedure Title	CM850 or CM552 Choice
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc442-choice (/qs3/pubsys2/xml/en/procedures/60/60-fc442-choice.html)
Procedure Title	CM850 or CM552 Choice
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc442-ecm1 (/qs3/pubsys2/xml/en/procedures/60/60-fc442-ecm1.html)
Procedure Title	Battery 1 Voltage – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc488 (/qs3/pubsys2/xml/en/procedures/60/60-fc488.html)
Procedure Title	Intake Manifold 1 Temperature – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc611 (/qs3/pubsys2/xml/en/procedures/60/60-fc611.html)
Procedure Title	Engine Shutdown Hot – Condition Exists
Update Reason	

Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc689 (/qs3/pubsys2/xml/en/procedures/60/60-fc689.html)
Procedure Title	Engine Crankshaft Speed – Data Erratic, Intermittent, or Incorrect
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1357 (/qs3/pubsys2/xml/en/procedures/60/60-fc1357.html)
Procedure Title	Engine Oil Level Remote Reservoir – Data Valid but Below Normal Operational Range – Moderately Severe Level
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1383 (/qs3/pubsys2/xml/en/procedures/60/60-fc1383.html)
Procedure Title	Intake Manifold 1 Pressure Sensor Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	09-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1891 (/qs3/pubsys2/xml/en/procedures/60/60-fc1891.html)
Procedure Title	Engine Oil Change Interval – Condition Exists
Update Reason	

Date	10-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc441-ecm1 (/qs3/pubsys2/xml/en/procedures/60/60-fc441-ecm1.html)
Procedure Title	Battery 1 Voltage – Data Valid but Below Normal Operational Range – Moderately Severe Level
Update Reason	

Date	10-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc442-ecm2 (/qs3/pubsys2/xml/en/procedures/60/60-fc442-ecm2.html)
Procedure Title	Battery 1 Voltage High – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	

Date	10-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1384 (/qs3/pubsys2/xml/en/procedures/60/60-fc1384.html)
Procedure Title	Intake Manifold 1 Pressure Sensor Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	10-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1386 (/qs3/pubsys2/xml/en/procedures/60/60-fc1386.html)
Procedure Title	Intake Manifold 2 Pressure Sensor Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1387 (/qs3/pubsys2/xml/en/procedures/60/60-fc1387.html)
Procedure Title	SAE J1939 Engine Commanded Shutdown – Condition Exists
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1411 (/qs3/pubsys2/xml/en/procedures/60/60-fc1411.html)
Procedure Title	Generator Output Frequency Adjust Potentiometer Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1412 (/qs3/pubsys2/xml/en/procedures/60/60-fc1412.html)
Procedure Title	Droop Adjust Potentiometer Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1418 (/qs3/pubsys2/xml/en/procedures/60/60-fc1418.html)
Procedure Title	Gain Adjust Potentiometer Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1517 (/qs3/pubsys2/xml/en/procedures/60/60-fc1517.html)
Procedure Title	At Least One Module Has A Most Severe Fault – Condition Exists
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1518 (/qs3/pubsys2/xml/en/procedures/60/60-fc1518.html)
Procedure Title	At Least One Module Has A Moderately Severe Fault – Condition Exists
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1519 (/qs3/pubsys2/xml/en/procedures/60/60-fc1519.html)
Procedure Title	At Least One Module Has A Least Severe Fault – Condition Exists
Update Reason	
Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1597 (/qs3/pubsys2/xml/en/procedures/60/60-fc1597.html)
Procedure Title	Engine Control Module Critical Internal Failure – Bad Intelligent Device or Component
Update Reason	

Date	15-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1847 (/qs3/pubsys2/xml/en/procedures/60/60-fc1847.html)
Procedure Title	Engine Coolant Temperature – Special Instructions
Update Reason	
Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc135 (/qs3/pubsys2/xml/en/procedures/60/60-fc135.html)
Procedure Title	Engine Oil Rifle Pressure 1 Sensor Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1978 (/qs3/pubsys2/xml/en/procedures/60/60-fc1978.html)
Procedure Title	Generator Speed/Load Governing Bias Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1979 (/qs3/pubsys2/xml/en/procedures/60/60-fc1979.html)
Procedure Title	Generator Speed/Load Governing Bias Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1984 (/qs3/pubsys2/xml/en/procedures/60/60-fc1984.html)
Procedure Title	Intake Manifold 2 Temperature – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1992 (/qs3/pubsys2/xml/en/procedures/60/60-fc1992.html)
Procedure Title	Engine Crankshaft Speed Sensor – Data Valid but Above Normal Operational Range – Moderately Severe Level
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2185 (/qs3/pubsys2/xml/en/procedures/60/60-fc2185.html)
Procedure Title	Sensor Supply 4 Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2186 (/qs3/pubsys2/xml/en/procedures/60/60-fc2186.html)
Procedure Title	Sensor Supply 4 Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2265 (/qs3/pubsys2/xml/en/procedures/60/60-fc2265.html)
Procedure Title	Electric Lift Pump for Engine Fuel Supply Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2266 (/qs3/pubsys2/xml/en/procedures/60/60-fc2266.html)
Procedure Title	Electric Lift Pump for Engine Fuel Supply Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2661 (/qs3/pubsys2/xml/en/procedures/60/60-fc2661.html)
Procedure Title	At Least One Unacknowledged Most Severe Fault – Condition Exists
Update Reason	

Date	17-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2662 (/qs3/pubsys2/xml/en/procedures/60/60-fc2662.html)
Procedure Title	At Least One Unacknowledged Moderately Severe Fault – Condition Exists
Update Reason	

Date	17-Jan-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	60-019-174 (/qs3/pubsys2/xml/en/procedures/60/60-019-174.html)
Procedure Title	Backbone Wiring Harness
Update Reason	
Date	17-Jan-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	60-019-363 (/qs3/pubsys2/xml/en/procedures/60/60-019-363.html)
Procedure Title	Camshaft Position Sensor
Update Reason	
Date	17-Jan-2008
Section	Back
Procedure Number	4021674-backcover (/qs3/pubsys2/xml/en/manual/4021674/4021674-backcover.html)
Procedure Title	Backcover
Update Reason	
Date	21-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1385 (/qs3/pubsys2/xml/en/procedures/60/60-fc1385.html)
Procedure Title	Intake Manifold 2 Pressure Sensor Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	

Date	23-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc234-ecm1 (/qs3/pubsys2/xml/en/procedures/60/60-fc234-ecm1.html)
Procedure Title	Engine Crankshaft Speed/Position – Data Valid but Above Normal Operational Range – Most Severe Level
Update Reason	
Date	23-Jan-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc2727 (/qs3/pubsys2/xml/en/procedures/60/60-fc2727.html)
Procedure Title	SAE J1939 Data Link 2 Engine Network – Abnormal Update Rate
Update Reason	
Date	28-Jan-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)
Procedure Title	Resistance Measurement Using a Multimeter
Update Reason	
Date	13-Feb-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc423 (/qs3/pubsys2/xml/en/procedures/60/60-fc423.html)
Procedure Title	Engine Fuel Timing Actuator Stuck – Data Erratic, Intermittent, or Incorrect
Update Reason	

Date	21-Feb-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-219 (/qs3/pubsys2/xml/en/procedures/99/99-019-219.html)
Procedure Title	Danfoss™ (Circular DIN Series) Connector
Update Reason	
Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t016-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t016-tr.html)
Procedure Title	Communication Error – Electronic Service Tool or Control Device
Update Reason	
Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t083-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t083-tr.html)
Procedure Title	Fault Code Warning Lamps Stay On (No Apparent Reason)
Update Reason	
Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t084-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t084-tr.html)
Procedure Title	Fault Code Warning Lamps Do Not Illuminate
Update Reason	

Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t087-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	
Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t091-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	
Date	27-Feb-2008
Section	Section TS – Troubleshooting Symptoms
Procedure Number	4021674-t092-tr (/qs3/pubsys2/xml/en/manual/4021674/4021674-t092-tr.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	
Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc173 (/qs3/pubsys2/xml/en/procedures/60/60-fc173.html)
Procedure Title	Fuel Rack Actuator Driver Mechanically Stuck Closed – Data Valid but Below Normal Operating Range – Moderately Severe Level
Update Reason	

Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc234-ecm2 (/qs3/pubsys2/xml/en/procedures/60/60-fc234-ecm2.html)
Procedure Title	Engine Speed High – Data Valid but Above Normal Operational Range – Most Severe Level
Update Reason	
Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc426 (/qs3/pubsys2/xml/en/procedures/60/60-fc426.html)
Procedure Title	SAE J1939 Data Link – Data Erratic, Intermittent, or Incorrect
Update Reason	
Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc441-ecm2 (/qs3/pubsys2/xml/en/procedures/60/60-fc441-ecm2.html)
Procedure Title	Battery 1 Voltage – Data Valid but Below Normal Operational Range – Moderately Severe Level
Update Reason	
Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1117 (/qs3/pubsys2/xml/en/procedures/60/60-fc1117.html)
Procedure Title	Power Supply Lost With Ignition On – Data Erratic, Intermittent, or Incorrect
Update Reason	

Date	28-Apr-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc1427 (/qs3/pubsys2/xml/en/procedures/60/60-fc1427.html)
Procedure Title	Overspeed Shutdown Relay Driver Diagnostic has Detected an Error – Condition Exists
Update Reason	
Date	05-May-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc213 (/qs3/pubsys2/xml/en/procedures/60/60-fc213.html)
Procedure Title	Engine Oil Temperature Sensor 1 Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	
Date	16-May-2008
Section	Front
Procedure Number	4021674-titlepage (/qs3/pubsys2/xml/en/manual/4021674/4021674-titlepage.html)
Procedure Title	Titlepage
Update Reason	
Date	21-May-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc254 (/qs3/pubsys2/xml/en/procedures/60/60-fc254.html)
Procedure Title	Engine Fuel Shutoff Valve Circuit – Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	30-May-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-028 (/qs3/pubsys2/xml/en/procedures/99/99-019-028.html)
Procedure Title	Diagnostic Test Mode Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-047 (/qs3/pubsys2/xml/en/procedures/99/99-019-047.html)
Procedure Title	Fault Lamp Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-071 (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)
Procedure Title	OEM Wiring Harness
Update Reason	

Date	03-Jun-2008
Section	Section L – Service Literature
Procedure Number	60-205-001 (/qs3/pubsys2/xml/en/procedures/60/60-205-001.html)
Procedure Title	Additional Service Literature
Update Reason	
Date	23-Jun-2008
Section	Section TF – Troubleshooting Fault Codes
Procedure Number	60-fc166 (/qs3/pubsys2/xml/en/procedures/60/60-fc166.html)
Procedure Title	Fuel Rack Position 1 Sensor Circuit – Voltage Above Normal or Shorted to High Source
Update Reason	
Date	09-Jul-2008
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	60-019-026 (/qs3/pubsys2/xml/en/procedures/60/60-019-026.html)
Procedure Title	Data Link Circuit, Public
Update Reason	
Date	19-Mar-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-214 (/qs3/pubsys2/xml/en/procedures/99/99-019-214.html)
Procedure Title	Bosch® Actuator and Sensor Connector Series
Update Reason	

Date	19-Mar-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-215 (/qs3/pubsys2/xml/en/procedures/99/99-019-215.html)
Procedure Title	Bosch® ECM Injector Driver Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-216 (/qs3/pubsys2/xml/en/procedures/99/99-019-216.html)
Procedure Title	Bosch® ECM Actuator and Sensor Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-217 (/qs3/pubsys2/xml/en/procedures/99/99-019-217.html)
Procedure Title	Bosch® ECM OEM Connector Series
Update Reason	
Date	09-Jun-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	60-019-042 (/qs3/pubsys2/xml/en/procedures/60/60-019-042.html)
Procedure Title	Engine Speed Sensor (ESS)
Update Reason	

Date	25-Jun-2009
Section	Section TS – Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L – Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	10-Sep-2009
Section	Section i – Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	
Date	16-Sep-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-064 (/qs3/pubsys2/xml/en/procedures/99/99-019-064.html)
Procedure Title	Key Switch Battery Supply Circuit
Update Reason	
Date	08-Oct-2009
Section	Section i – Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	

Date	20-Oct-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	105-019-427 (/qs3/pubsys2/xml/en/procedures/105/105-019-427.html)
Procedure Title	ECM ROM Boot
Update Reason	
Date	02-Dec-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	
Date	02-Dec-2009
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	
Date	25-Jan-2010
Section	Section i – Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	

Date	25-Jan-2010
Section	Section V – Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V – Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	24-Mar-2010
Section	Section i – Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	10-May-2010
Section	Section V – Specifications
Procedure Number	4021683 (/qs3/pubsys2/xml/en/procedures//4021683.html)
Procedure Title	Specifications
Update Reason	
Date	25-May-2010
Section	Section i – Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	

Date	16-Jun-2010
Section	Section 19 - Electronic Engine Controls – Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	
Date	17-Aug-2010
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-474 (/qs3/pubsys2/xml/en/procedures/99/99-019-474.html)
Procedure Title	Bosch Aftertreatment DEF Controller Connector
Update Reason	Added procedure to outline.
Date	29-Oct-2010
Section	Section 19 – Electronic Engine Controls – Group 19
Procedure Number	60-019-043 (/qs3/pubsys2/xml/en/procedures/60/60-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	Added a Resistance Check section.
Date	10-Feb-2012
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	Updated to include information concerning troubleshooting of ECM calibration issues once new calibration is installed.

Date	17-Apr-2014
Section	Section 19 - Electronic Controls
Procedure Number	60-019-031 (/qs3/pubsys2/xml/en/procedures/60/60-019-031.html)
Procedure Title	Engine Control Module
Update Reason	Revise jackscrew torque value and added a note to not over torque.

Date	17-Apr-2014
Section	Section 19 - Electronic Controls
Procedure Number	60-019-043 (/qs3/pubsys2/xml/en/procedures/60/60-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	Revise jackscrew torque value and added a note to not over torque.

Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.

Date	12-Dec-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	60-t05-423 (/qs3/pubsys2/xml/en/procedures/60/60-t05-423.html)
Procedure Title	Engine Fuel Timing Actuator Stuck - Data Erratic, Intermittent, or Incorrect
Update Reason	Changed Guided Step 4A, Specifications from Fault Codes 234 to 432.

Date	14-Aug-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	60-t05-173 (/qs3/pubsys2/xml/en/procedures/60/60-t05-173.html)
Procedure Title	Fuel Rack Actuator Driver Mechanically Stuck Closed - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	Revised Step 2A to include escalation directions.

Date	04-Apr-2019
Section	Section 19 - Electronic Controls
Procedure Number	60-019-397 (/qs3/pubsys2/xml/en/procedures/60/60-019-397.html)
Procedure Title	Fuel Lift Pump Circuit
Update Reason	Updated procedure to include connector required for voltage check during Key-On.

Date	21-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	Engine Control Module Calibration Code
Update Reason	Added the following statement to the Preparatory Steps of the procedure: "Verify vehicle odometer is equal to the value recorded before ECM removal. Contact OEM service location if values are incorrect."

Date	12-Feb-2021
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	60-019-397 (/qs3/pubsys2/xml/en/procedures/60/60-019-397.html)
Procedure Title	Fuel Lift Pump Circuit
Update Reason	Added Cable, Electrical Breakout Harness, Part Number 3163531 to Additional Service Items list. Reorganized Resistance Check steps. Rewrote Voltage Check steps.

